

Lecture 11 (10/9)

Autoregressive models, regression to the mean, and the Hurwicz bias

- More practice working with (single-predictor) regression models
- Time series models (the fun part)

p -values vs. R^2

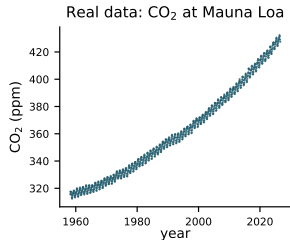
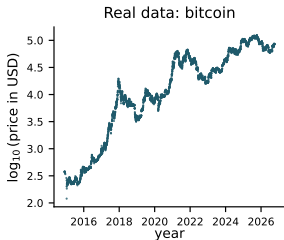
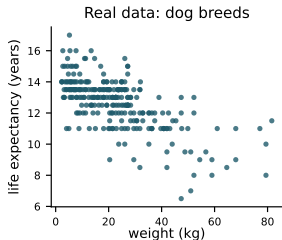
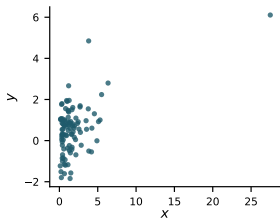
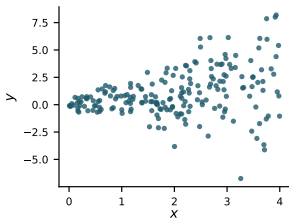
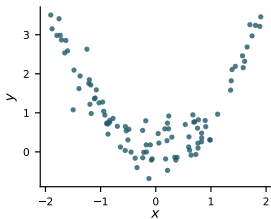
Question:

Is each of the following possible? If so, how? (The p -value is for the null hypothesis $H_0 : \beta_1 = 0$.)

- A very small p -value, but $R^2 \approx 0$.
- A p -value $\approx 1/2$, but R^2 close to 1.
- A massive β_1 but tiny R^2 .

p -values vs. R^2

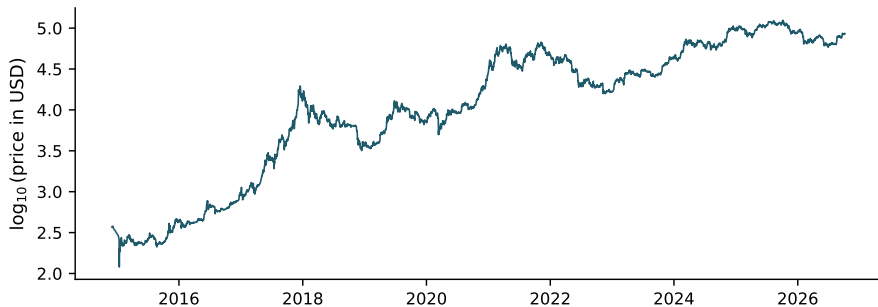
Guess R^2 and the p -value for $H_0 : \beta_1 = 0$ in each plot. No calculations needed!



Time series data

The price of bitcoin each day since December 2014, on a log scale:

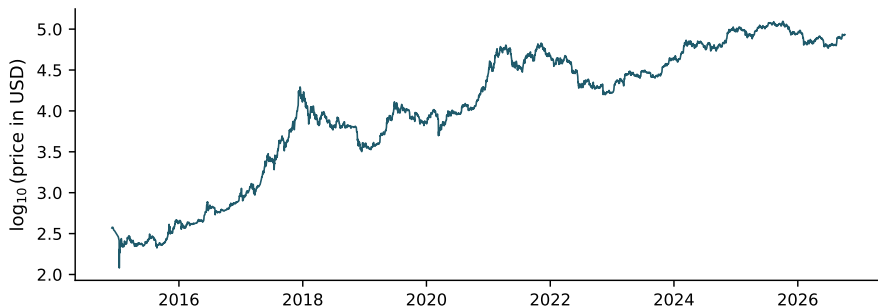
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How do we model this?

Conditioning on the history

In general, the next value depends on the whole history:

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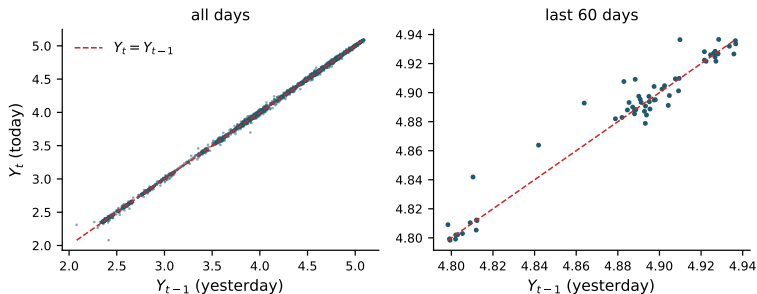
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Autoregressive models

An autoregressive model (AR(1)) is a linear regression model where the predictor is the previous value of the response:

$$Y_t \mid Y_{t-1} \sim \text{Normal}(\beta_0 + \beta_1 Y_{t-1}, \sigma_\epsilon^2), \quad \text{i.e.} \quad Y_t = \beta_0 + \beta_1 Y_{t-1} + \epsilon_t.$$

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The data for the regression are the pairs

$$(X, Y) = (Y_1, Y_2), (Y_2, Y_3), \dots, (Y_{T-1}, Y_T),$$

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Example

For bitcoin, least squares gives $\hat{\beta}_1 \approx 1.000$. What does this say about tomorrow's value?

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Y_t and Y_{t-1} are the *same quantity*, with the same units, so β_1 must be dimensionless.

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Suppose the series is stationary: Y_t and Y_{t-1} have the same mean μ and the same standard deviation σ_Y . Then

$$\beta_1 = \frac{\text{cov}(Y_{t-1}, Y_t)}{\sigma_Y^2} = \frac{\text{cov}(Y_{t-1}, Y_t)}{\sigma_Y \sigma_Y} = \rho,$$

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If $|\rho| < 1$, each step shrinks the deviation from the mean by a factor ρ :
regression to the mean.

Regression to the mean: some history

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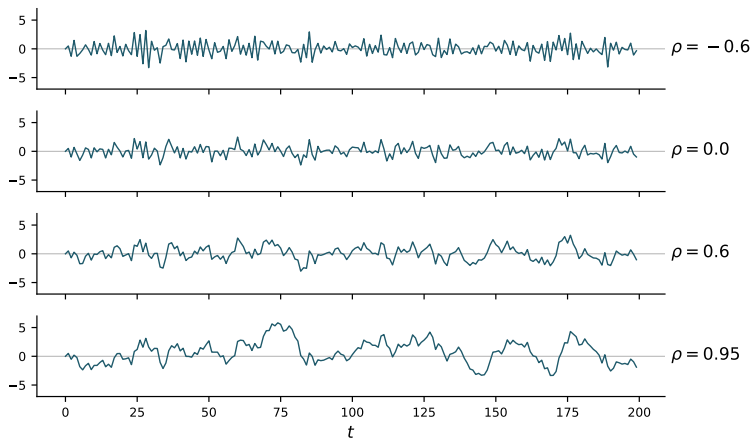
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- It shows up everywhere: the “sophomore slump” after a great rookie season, or flight instructors who conclude that praise makes pilots worse and criticism makes them better (Kahneman).

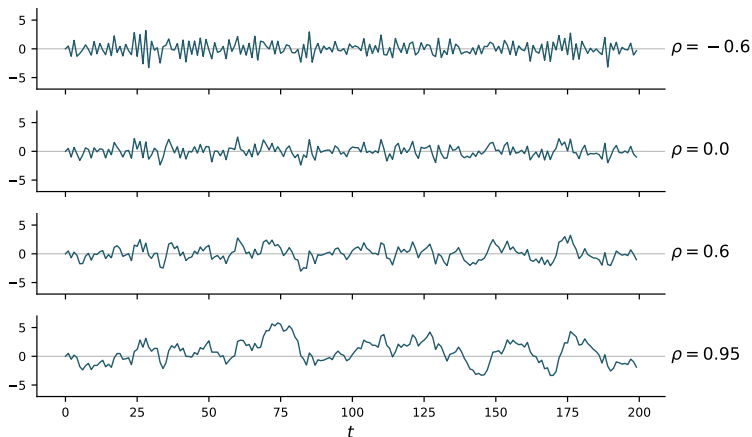
Simulations with different ρ

$Y_t = \rho Y_{t-1} + \epsilon_t$ with $\epsilon_t \sim \text{Normal}(0, 1)$, the same noise in every panel:



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Larger ρ : longer excursions away from the mean. Negative ρ : the series flips sign from step to step.

The Hurwicz bias

Take $\mu = 0$ and fit $Y_t = \rho Y_{t-1} + \epsilon_t$ by least squares (no intercept):

$$\hat{\rho} = \frac{\sum_t Y_{t-1} Y_t}{\sum_t Y_{t-1}^2}$$

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Here the predictors are $Y_1, \dots, Y_{T-1}$ themselves. ϵ_t is independent of the past (Y_{t-1}), but it is part of every future value:

$$Y_t = \rho Y_{t-1} + \epsilon_t, \quad Y_{t+1} = \rho Y_t + \epsilon_{t+1}, \quad \dots$$

So the predictors are correlated with the noise, and we can no longer argue the second term averages to 0.

The Hurwicz bias

Why is it negative? Each term $Y_{t-1}\epsilon_t$ in the numerator has mean 0, but the numerator and denominator move together:

- when ϵ_t has the same sign as Y_{t-1} , the numerator goes up *and* $|Y_t|$ is large, which makes the denominator $\sum_t Y_{t-1}^2$ large;
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A longer calculation (Hurwicz, 1950) gives, for large T ,

$$E[\hat{\rho}] - \rho \approx -\frac{2\rho}{T}.$$

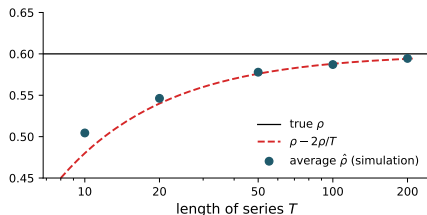
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Biased toward 0 at every T , but consistent.

How fast do correlations decay?

The autocorrelation at lag k is $\rho_k = \text{corr}(Y_t, Y_{t+k})$. For AR(1), iterating $Y_{t+1} = \rho Y_t + \epsilon_{t+1}$ gives

$$Y_{t+k} = \rho^k Y_t + (\text{noise independent of } Y_t) \quad \implies \quad \rho_k = \rho^k.$$

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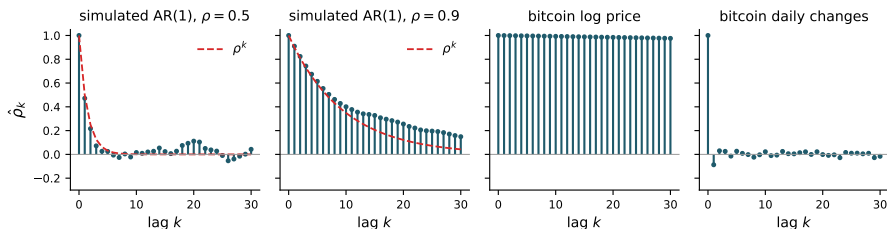
Sample version: $\hat{\rho}_k = \frac{\sum_t (Y_t - \bar{Y})(Y_{t+k} - \bar{Y})}{\sum_t (Y_t - \bar{Y})^2}.$

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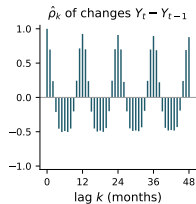
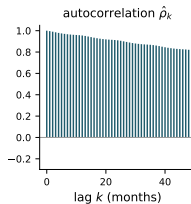
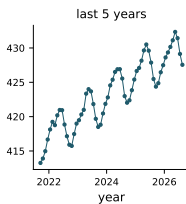
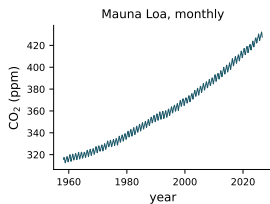
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The log price itself stays correlated for a long time; its daily changes $Y_t - Y_{t-1}$ are almost uncorrelated.

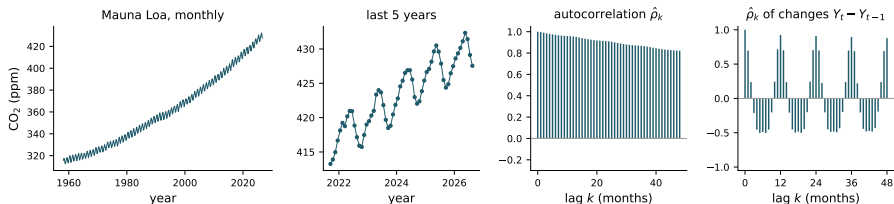
Climate data

Monthly CO₂ concentration at Mauna Loa, Hawaii (NOAA):



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- Is an AR(1) model reasonable here? What does $\hat{\rho}_k \approx 1$ for every k tell you?
- What does the autocorrelation of the month-to-month changes show?
- In Assignment 2 you will make the same kind of plot for daily casualty counts. A series with a strong trend (like a cumulative count) is strongly correlated with time and with itself, whatever the day-to-day details.